

LAB 01

GitHub Identity, Student Portal Registration,
and Ubuntu Server Installation

NAME

Maleeha Nadeem

REGISTRATION NUMBER

2024-BSE-13

COURSE CODE

SE-665

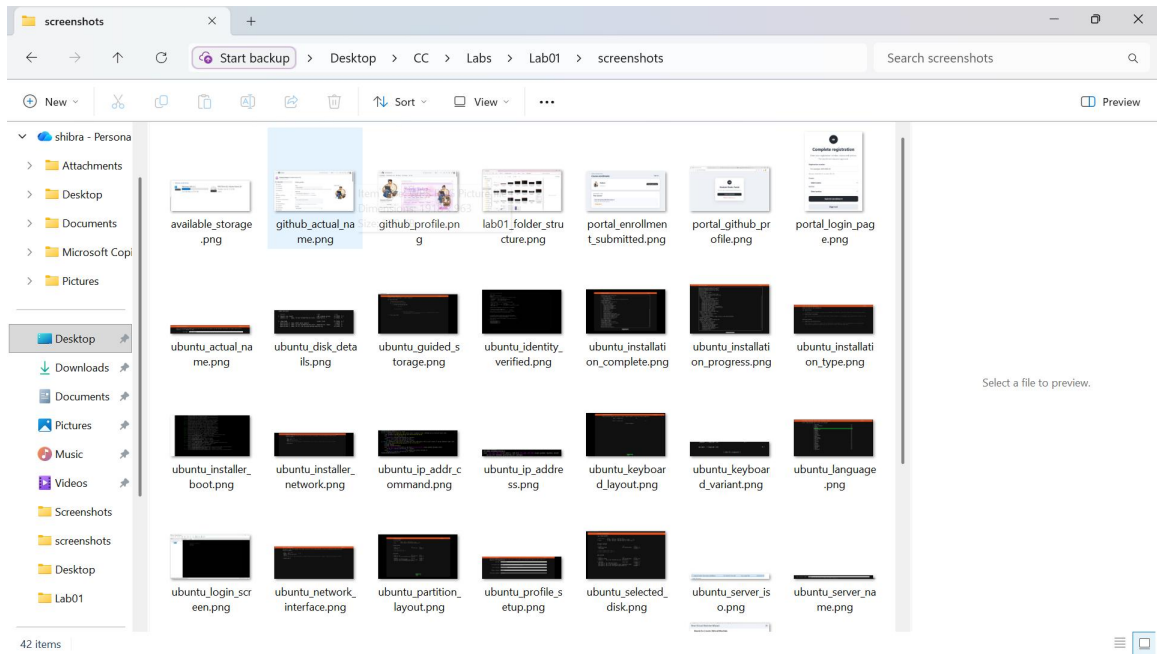
SECTION

A

GITHUB USERNAME

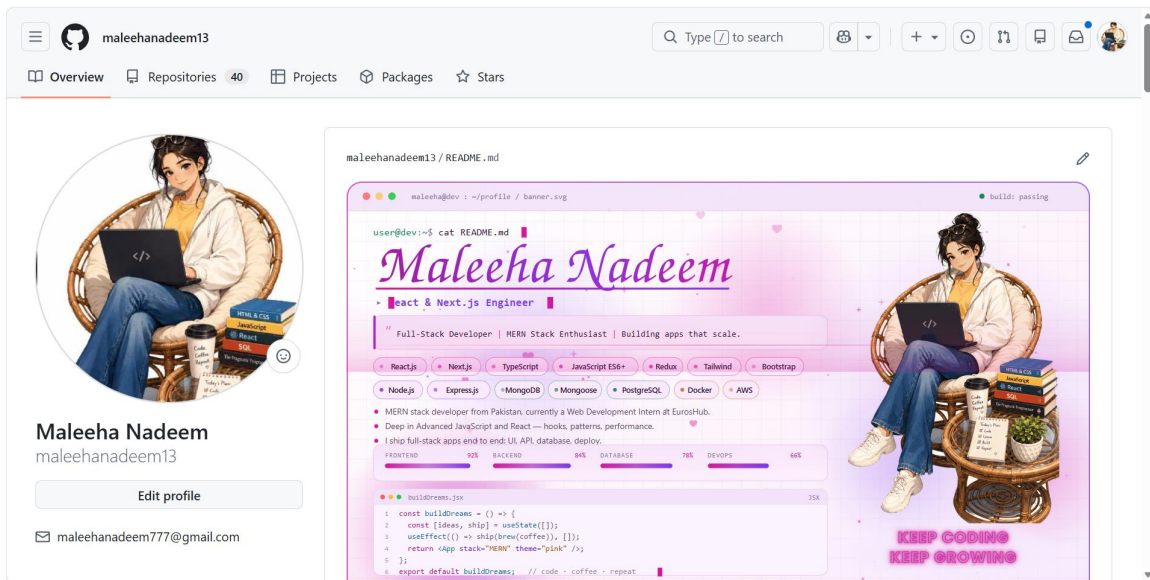
maleehanadeem13

Getting Started

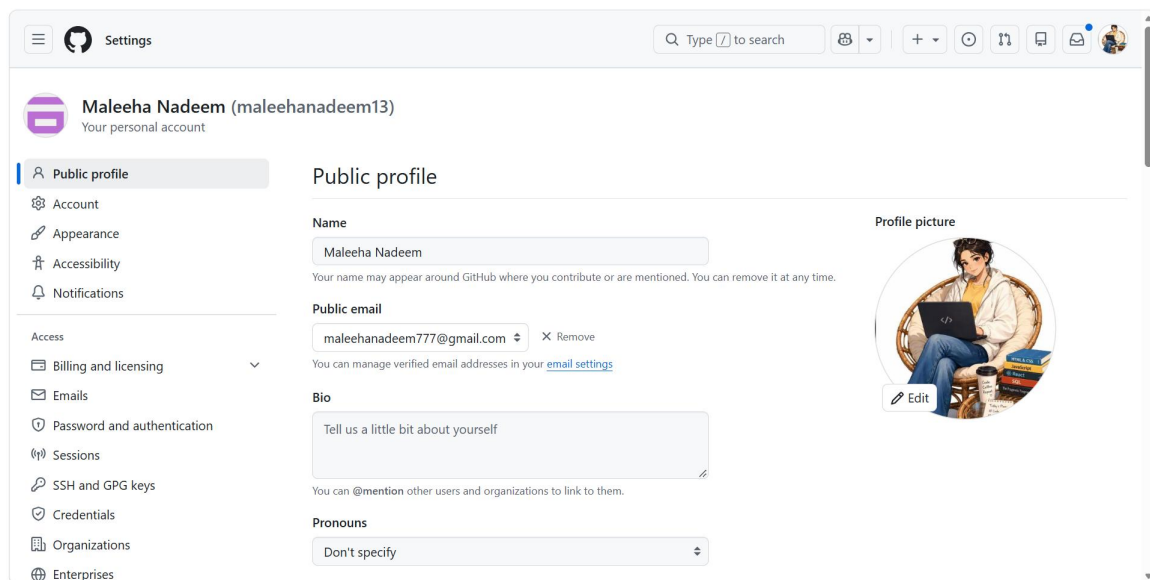


Created directory structure Labs/Lab01/screenshots/ inside the CC repository, ready to store evidence for this lab.

Task 1: Create and Configure Your GitHub Account

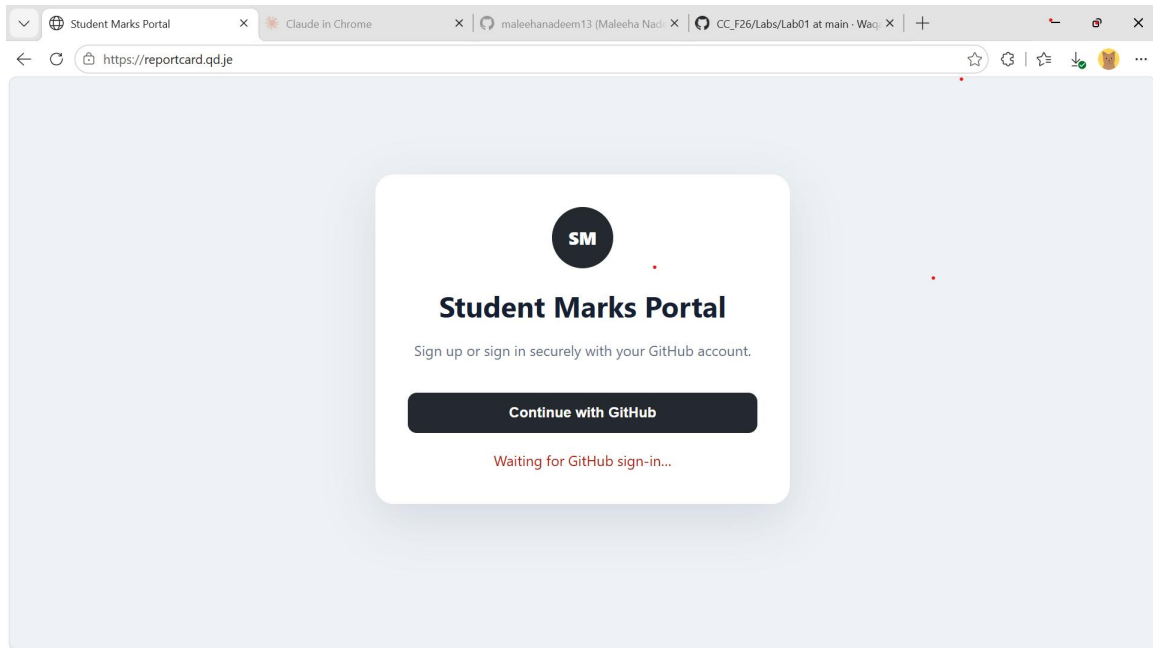


GitHub profile overview page showing the account name 'Maleeha Nadeem' and username 'maleehanadeem13'.



GitHub Settings > Public profile page with the Name field set to 'Maleeha Nadeem', confirming the complete actual name is on the account.

Task 2: Register in the Student Marks Portal



Student Marks Portal login page, prompting sign-in with GitHub.



Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

For example: 2024-BSE-01

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

Select course



Section

Select section



Submit enrollment

Sign out

Student Marks Portal registration form for entering the registration number, course, and section after signing in with GitHub.

Course enrollments

Sign out



Student

2024-BSE-13

+ Enroll in another course

ENROLLMENT STATUS

Your courses

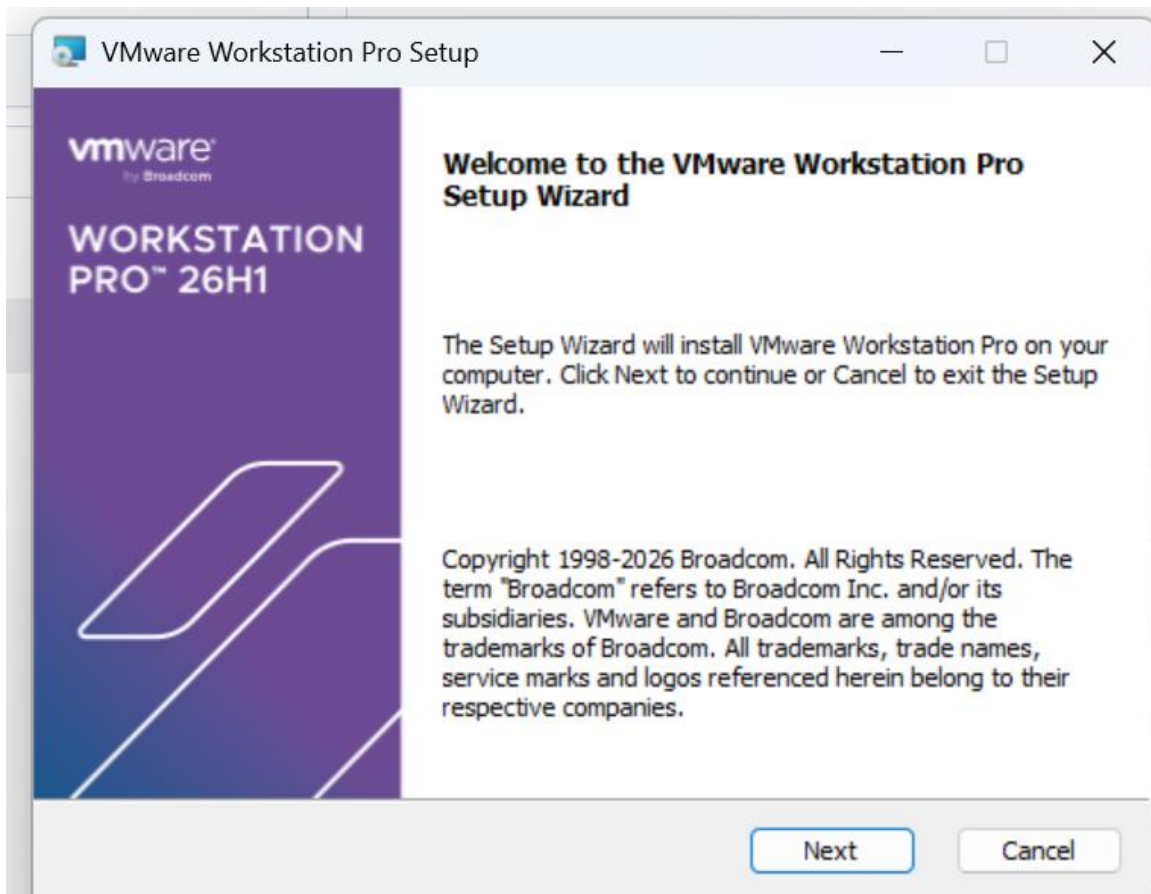
Cloud Computing (SE-665) Section A

Waiting for administrator approval

Pending approval

Course enrollment submitted for Cloud Computing (SE-665), Section A, under registration number 2024-BSE-13, showing status 'Pending approval'.

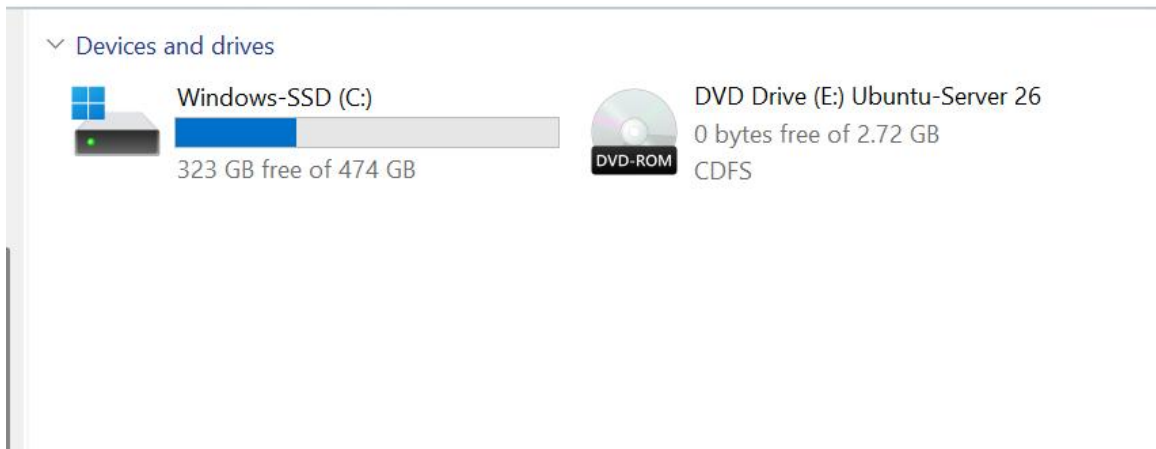
Task 3: Download VMware Workstation Pro and Ubuntu Server



VMware Workstation Pro running on Windows, confirming the hypervisor is installed and functional.

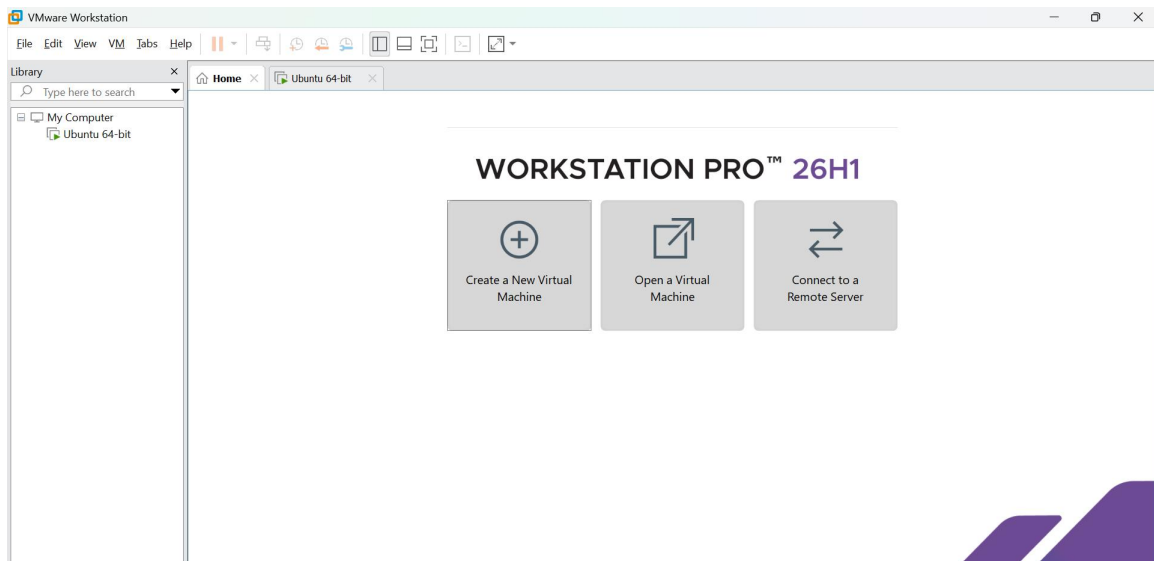
Files and folders			
Downloaded files			
 ubuntu-26.04.1-live-server-amd64.iso	9/17/2026 11:44 AM	Disc Image File	2,859,240 KB
▼ Earlier this week			

Downloaded Ubuntu Server LTS ISO file, confirming the installation media is ready for use.



Available disk storage on the host machine, confirming at least 20 GB of free space for the Ubuntu Server virtual machine.

Task 4: Create the Ubuntu Server Virtual Machine



New Virtual Machine Wizard opened in VMware Workstation Pro to begin creating the Ubuntu Server VM.

New Virtual Machine Wizard


WORKSTATION
PRO™ 26H1

Welcome to the New Virtual Machine Wizard

What type of configuration do you want?

☒ Typical (recommended)
Create a Workstation 25H2 or later virtual machine in a few easy steps.

☐ Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.

Help

< Back

Next >

Cancel

'Typical (recommended)' configuration selected for the new virtual machine.

New Virtual Machine Wizard

✕

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

 DVD Drive (E:) Ubuntu-Server 26

▼

☒ Installer disc image file (iso):

C:\Users\User\Downloads\ubuntu-26.04.1-live-server-a

▼

Browse...

Reading disc...

☐ I will install the operating system later.

The virtual machine will be created with a blank hard disk.

Help

< Back

Next >

Cancel

'Installer disc image file (ISO)' option selected with the downloaded Ubuntu Server ISO path chosen.

New Virtual Machine Wizard

✕

Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Ubuntu 64-bit.

The virtual machine will be created with the following settings:

Name:	Ubuntu 64-bit
Location:	C:\Users\User\Documents\Virtual Machines\Ubuntu 64-bit
Version:	Workstation 25H2 or later
Operating System:	Ubuntu 64-bit
Hard Disk:	20 GB, Split
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	2 CPU cores, CD/DVD, USB Controller, Sound Card

Customize Hardware...

☒ Power on this virtual machine after creation

< Back

Finish

Cancel

Final virtual machine configuration summary showing the allocated disk space and settings before creation.

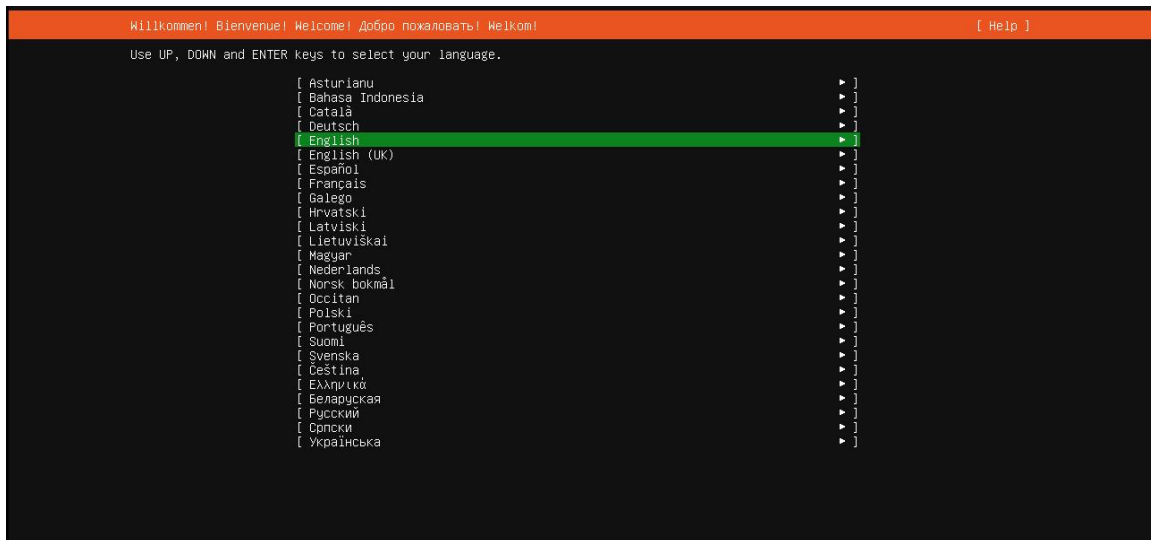
```

Starting system-tmpfiles-setup.service - Create System Files and Directories...
Starting ufw.service - Uncomplicated Firewall...
[ OK ] Finished finalrd.service - Create _runtime dir for shutdown pivot root.
[ OK ] Finished plymouth-read-write.service - Plymouth To Write Out Runtime Data.
[ OK ] Finished ufw.service - Uncomplicated Firewall.
Mounting proc-sys-fs-binfmt_misc.mount - Mountable File Formats File System...
[ OK ] Mounted proc-sys-fs-binfmt_misc.mount - Mountable File Formats File System.
[ OK ] Finished systemd-binfmt.service - Set Up Additional Binary Formats.
[ OK ] Finished systemd-udev-trigger.service - Coldplug All udev Devices.
[ OK ] Started systemd-ask-password-console-requests.service - Console Directory Watch.
[ OK ] Reached target cryptsetup.target - Local Encrypted Volumes.
[ OK ] Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
Starting ldconfig.service - Rebuild Dynamic Linker Cache...
Starting systemd-journal-catalog.service - Rebuild Journal Catalog...
[ OK ] Started vgauth.service - Authentic. virtual machines hosted on VMware.
[ OK ] Started open-vm-tools.service - Set up virtual machines hosted on VMware.
[ OK ] Finished kdump-tools.service - Kernel crash dump capture service.
[ OK ] Finished systemd-journal-catalog.service - Rebuild Journal Catalog.
[ OK ] Finished netplan-config.service - Netplan Backend Configuration.
[ OK ] Started snapd.apparmor.service - Snapfiles managed internally by snapd.
[ OK ] Finished console-setup.service - Set console font and keymap.
[ OK ] Listening on systemd-rfkill.socket - Load/Save RF Kill Switch Status /dev/rfkill Watch.
[ OK ] Started cloud-init-main.service - Cloud-Init: Single Process.
Starting cloud-init-local.service - Cloud-Init: Local Stage (pre-network)...
[ OK ] Reached target sound.target - Sound Card.
[ OK ] Finished ldconfig.service - Rebuild Dynamic Linker Cache.
Starting systemd-update-done.service - Update is Completed...
[ OK ] Finished systemd-update-done.service - Update is Completed.
[ OK ] Finished cloud-init-local.service - Cloud-Init: Local Stage (pre-network).
[ OK ] Reached target network-pre.target - Preparation for Network.
[ OK ] Listening on systemd-networkd-resolve-hook.socket - Network Management Resolve Hook Socket.
Starting cloud-init-network.service - Cloud-Init: Network Stage...
Starting systemd-networkd.service - Network Management...
[ OK ] Started systemd-networkd.service - Network Management.
[ OK ] Reached target network.target - Network.
Starting systemd-networkd-persistent-storage.service - Enable Persistent Storage in systemd-networkd...
Starting systemd-networkd-wait-online.service - Wait for Network to be Online...
[ OK ] Finished systemd-networkd-wait-online.service - Wait for Network to be Online.
[ OK ] Finished systemd-networkd-persistent-storage.service - Enable Persistent Storage in systemd-networkd.

```

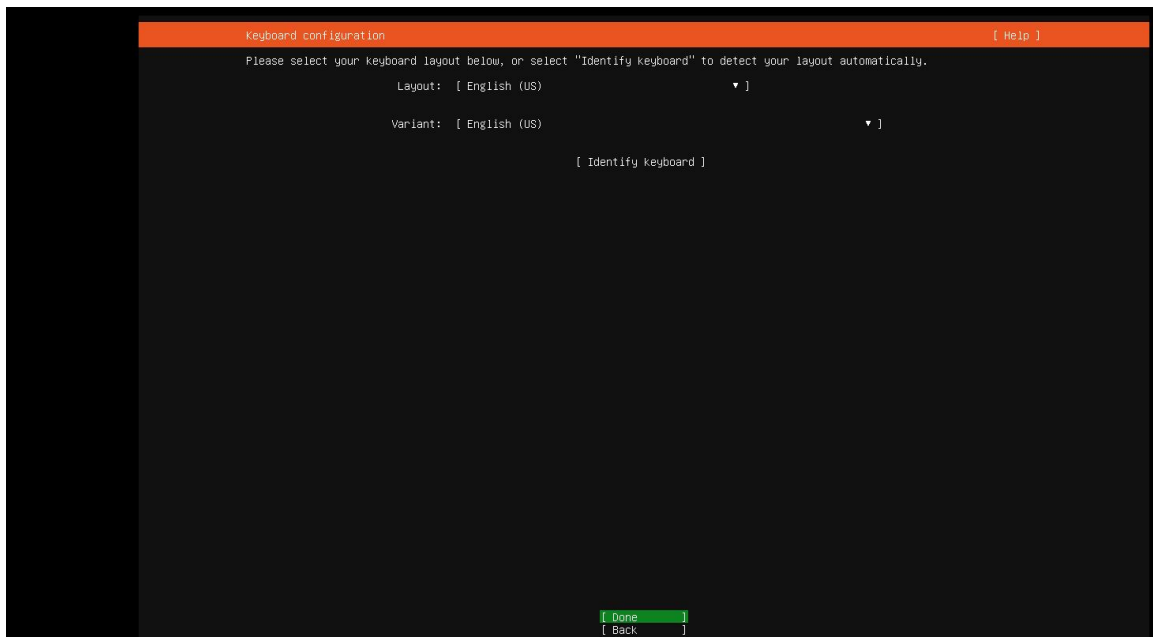
Ubuntu Server boot screen displayed after powering on the newly created virtual machine.

Task 5: Choose the Installation Language

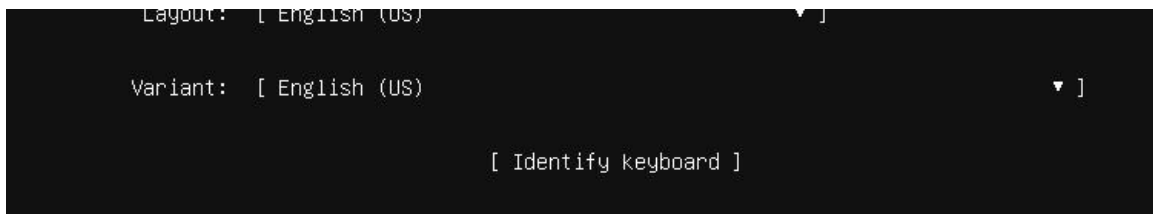


Installer language selection screen with the required installation language highlighted.

Task 6: Configure the Keyboard Layout



Keyboard layout selection screen showing the layout matching the host keyboard.



Keyboard variant selection confirming the correct variant for the chosen layout.

Task 7: Choose the Ubuntu Server Installation Type

Choose the type of installation

[Help]

Choose the base for the installation.

☒ Ubuntu Server

The default install contains a curated set of packages that provide a comfortable experience for operating your server.

☐ Ubuntu Server (minimized)

This version has been customized to have a small runtime footprint in environments where humans are not expected to log in.

Additional options

☐ Search for third-party drivers

This software is subject to license terms included with its documentation. Some is proprietary. Third-party drivers should not be installed on systems that will be used for FIPS or the real-time kernel.

Normal Ubuntu Server installation option selected (not MAAS).

Task 8: Configure the Network

```
Network configuration [ Help ]

Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient
access for updates.

NAME    TYPE    NOTES
[ ens33  eth    -          ▶ ]
DHCPv4   192.168.148.128/24
DHCPv6   -
00:0c:29:35:9b:1d / Intel Corporation / 82545EM Gigabit Ethernet Controller (Copper) (PRO/1000 MT Single Port Adapter)

[ Create bond ▶ ]
```

Network interface detected by the Ubuntu Server installer.

```
Network configuration [ Help ]

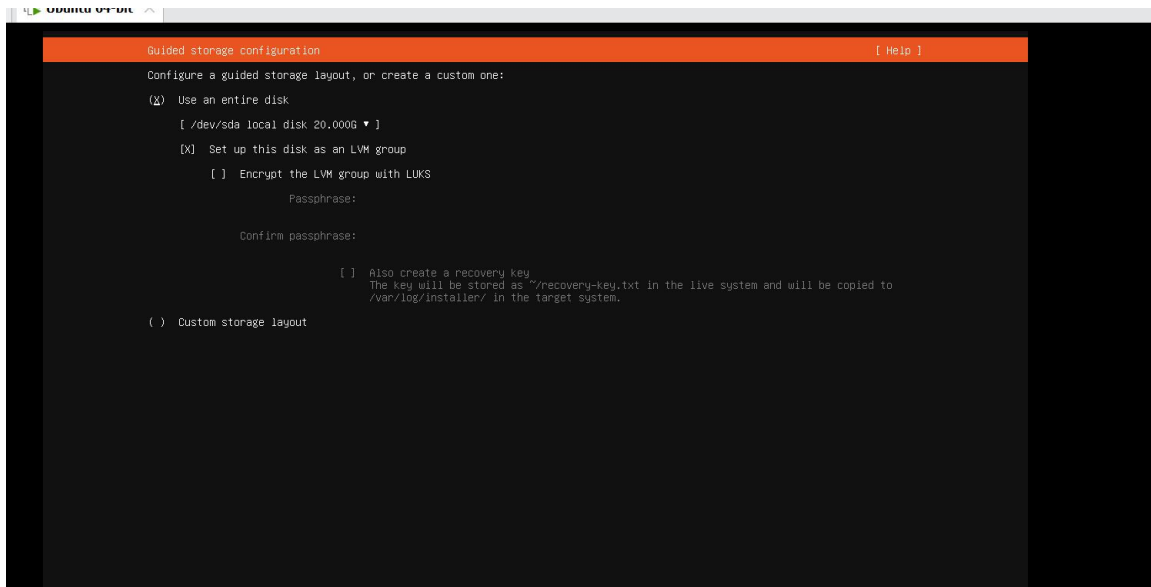
Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient
access for updates.

NAME    TYPE    NOTES
[ ens33  eth    -          ▶ ]
DHCPv4   192.168.148.128/24
DHCPv6   -
00:0c:29:35:9b:1d / Intel Corporation / 82545EM Gigabit Ethernet Controller (Copper) (PRO/1000 MT Single Port Adapter)

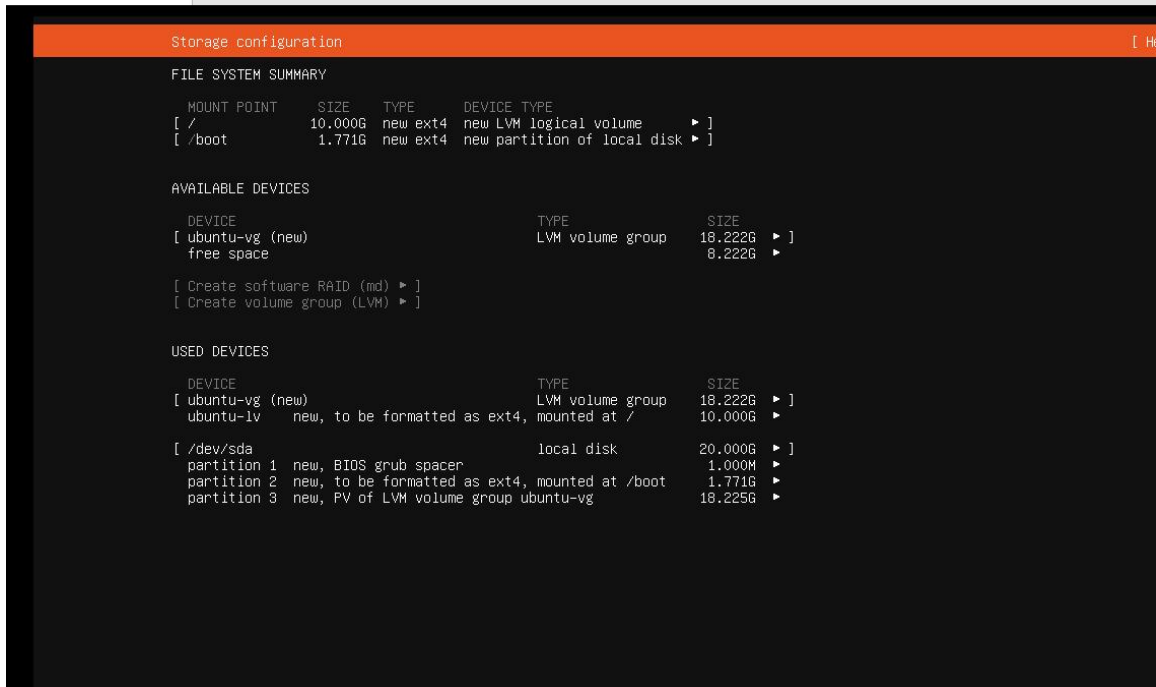
[ Create bond ▶ ]
```

Network interface showing an IP address automatically assigned via DHCP.

Task 9: Configure Storage

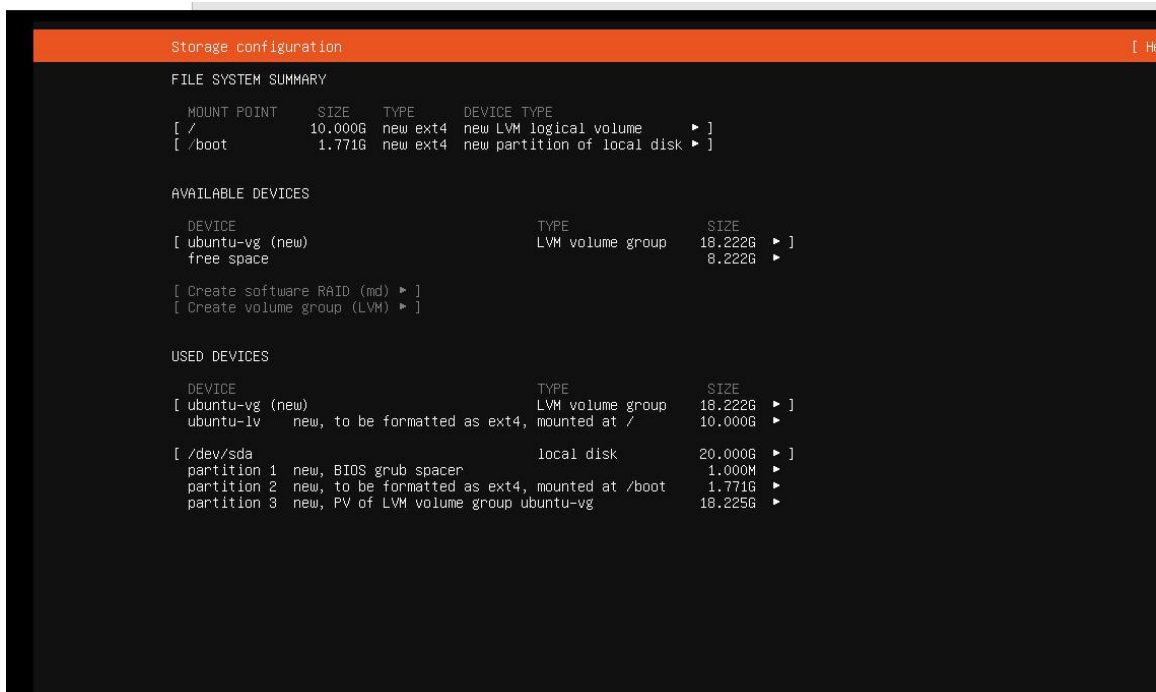


Guided storage configuration option selected to use the VMware virtual disk.

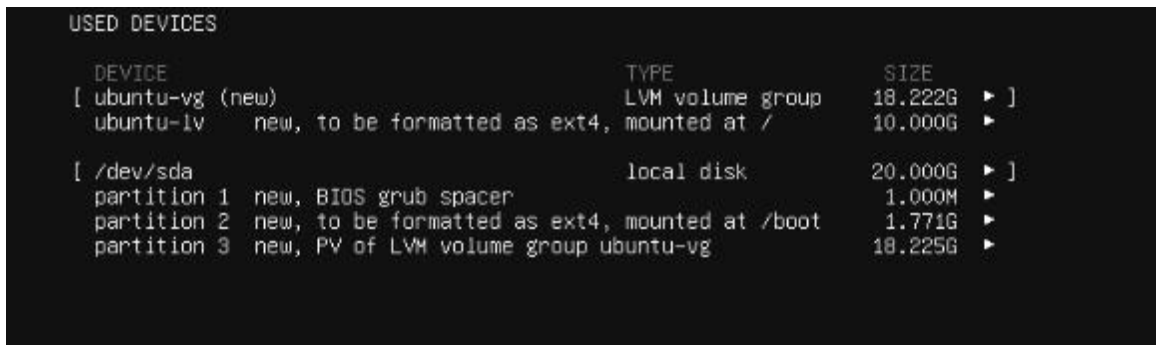


Storage configuration screen reviewed before proceeding with partitioning.

Task 10: Select the Installation Disk

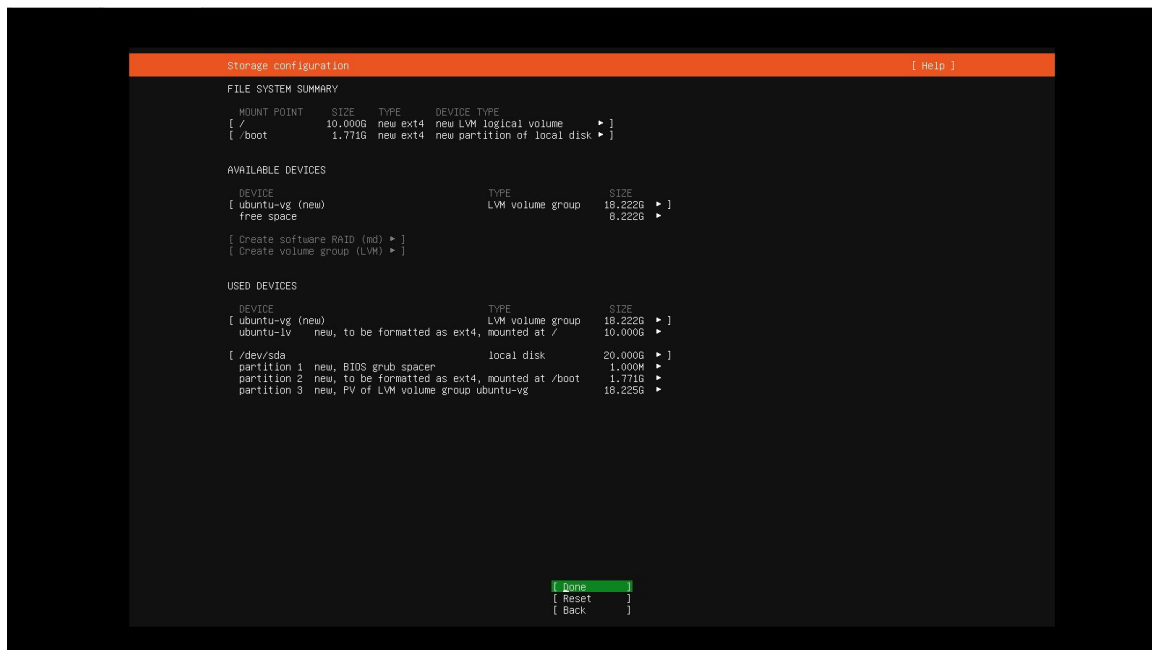


Virtual disk created for the Ubuntu Server VM selected as the installation target.



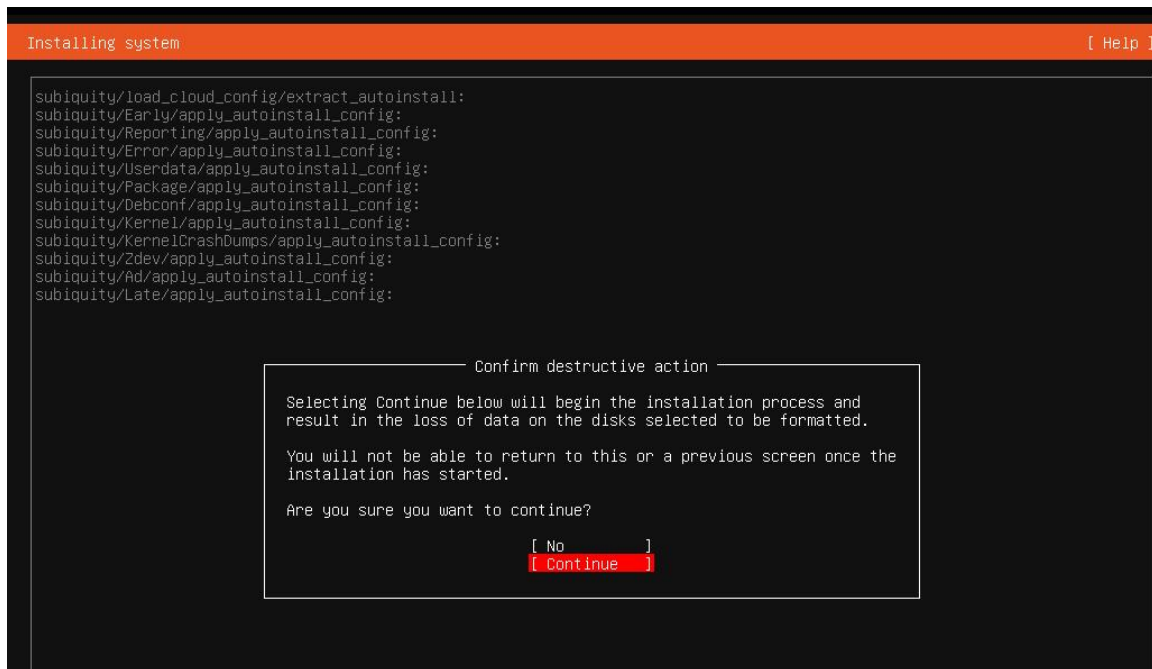
Selected disk's name and size confirmed as the correct VMware virtual disk.

Task 11: Review the Partitions

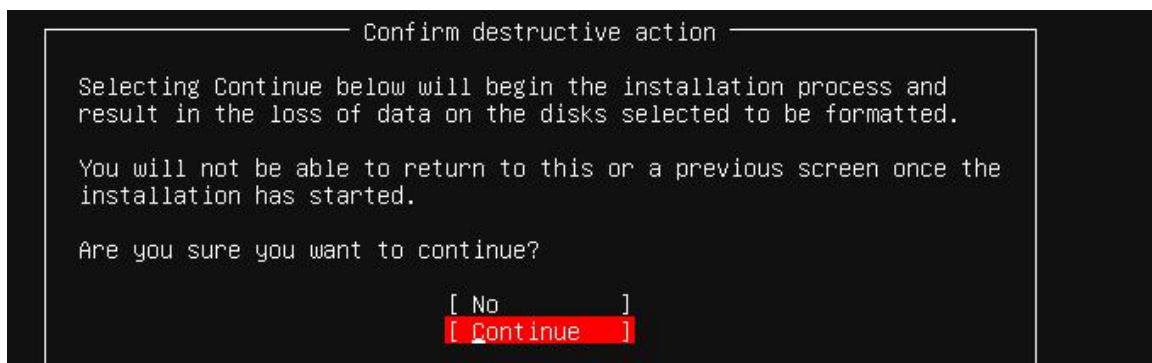


Proposed partition layout on the VMware virtual disk reviewed before continuing.

Task 12: Confirm the Storage Changes



Final destructive-action warning displayed before storage changes are applied.



Storage changes confirmed on the correct VMware virtual disk.

Task 13: Configure the Ubuntu Profile



Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

'Your name' field on the profile setup screen filled in with 'Maleeha Nadeem'.



Your servers name:

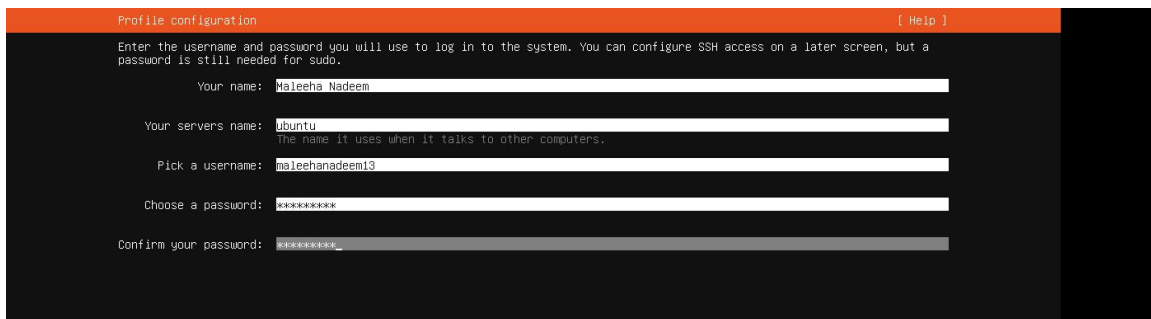
The name it uses when it talks to other computers.

'Your server's name' field set to 'ubuntu' as required.



Pick a username:

'Pick a username' field set to 'maleehanadeem13', matching the GitHub username exactly.



Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:

The name it uses when it talks to other computers.

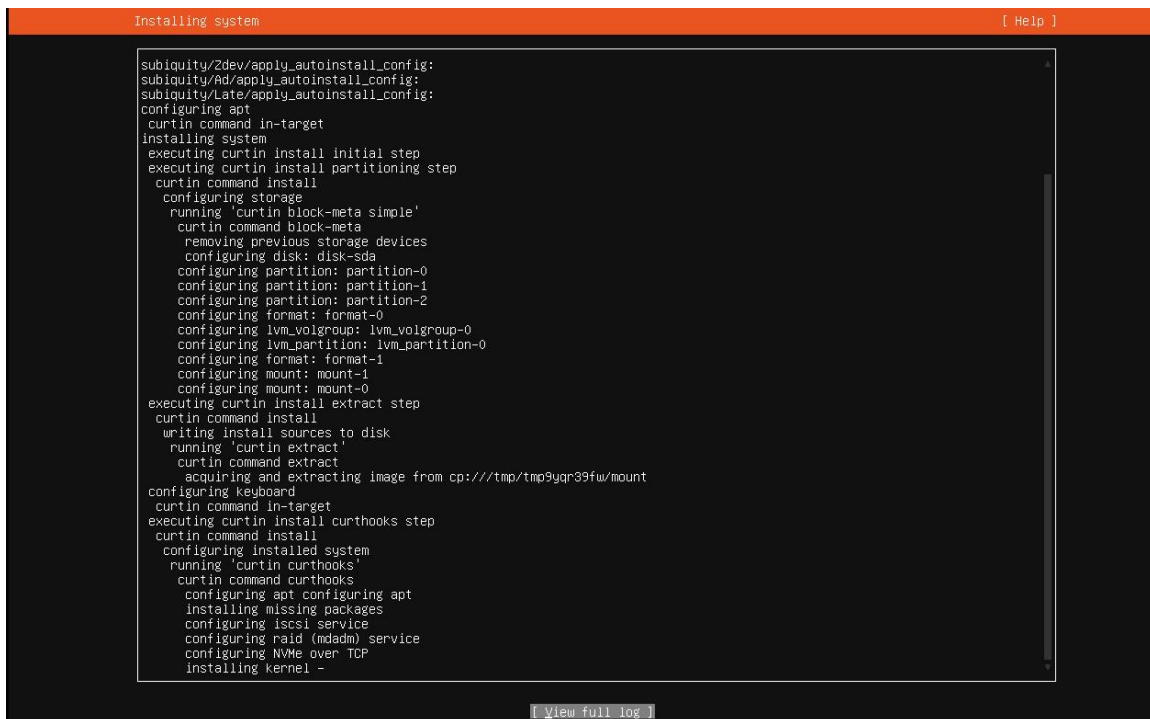
Pick a username:

Choose a password:

Confirm your password:

Completed Ubuntu profile setup page reviewed before continuing the installation.

Task 14: Complete the Software Installation



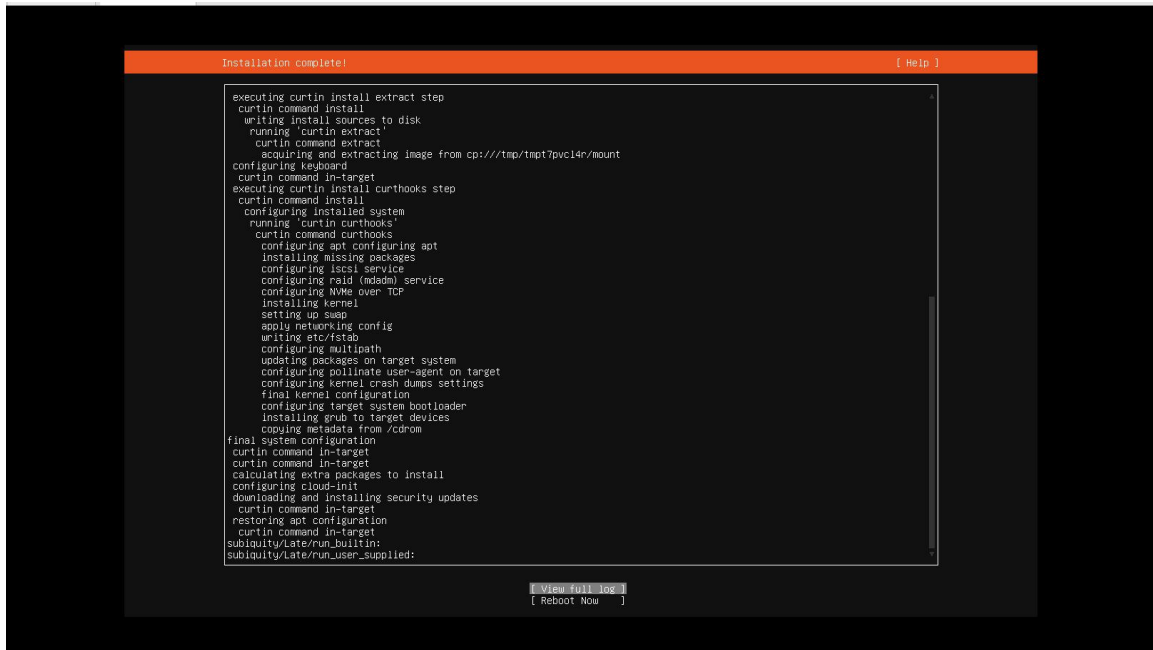
The screenshot shows the Ubuntu Server installation progress screen. The title bar at the top is orange and contains the text "Installing system" on the left and "[Help]" on the right. The main content area is a dark gray terminal window displaying the installation progress. The text is as follows:

```
subiquity/2dev/apply_autoInstall_config:
subiquity/Ad/apply_autoInstall_config:
subiquity/Late/apply_autoInstall_config:
configuring apt
  curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
  curtin command install
    configuring storage
      running 'curtin block-meta simple'
      curtin command block-meta
      removing previous storage devices
      configuring disk: disk-sda
      configuring partition: partition-0
      configuring partition: partition-1
      configuring partition: partition-2
      configuring format: format-0
      configuring lvm_volgroup: lvm_volgroup-0
      configuring lvm_partition: lvm_partition-0
      configuring format: format-1
      configuring mount: mount-1
      configuring mount: mount-0
    executing curtin install extract step
      curtin command install
      uniting install sources to disk
      running 'curtin extract'
      curtin command extract
      acquiring and extracting image from cp:///tmp/tmp9yqr39fw/mount
    configuring keyboard
  curtin command in-target
executing curtin install curthooks step
  curtin command install
    configuring installed system
      running 'curtin curthooks'
      curtin command curthooks
      configuring apt configuring apt
      installing missing packages
      configuring iscsi service
      configuring raid (mdadm) service
      configuring NVMe over TCP
      installing kernel -
```

At the bottom of the terminal window, there is a button labeled "View full log".

Ubuntu Server installation progress screen while the required software installs.

Task 15: Reboot and Verify the Ubuntu Identity

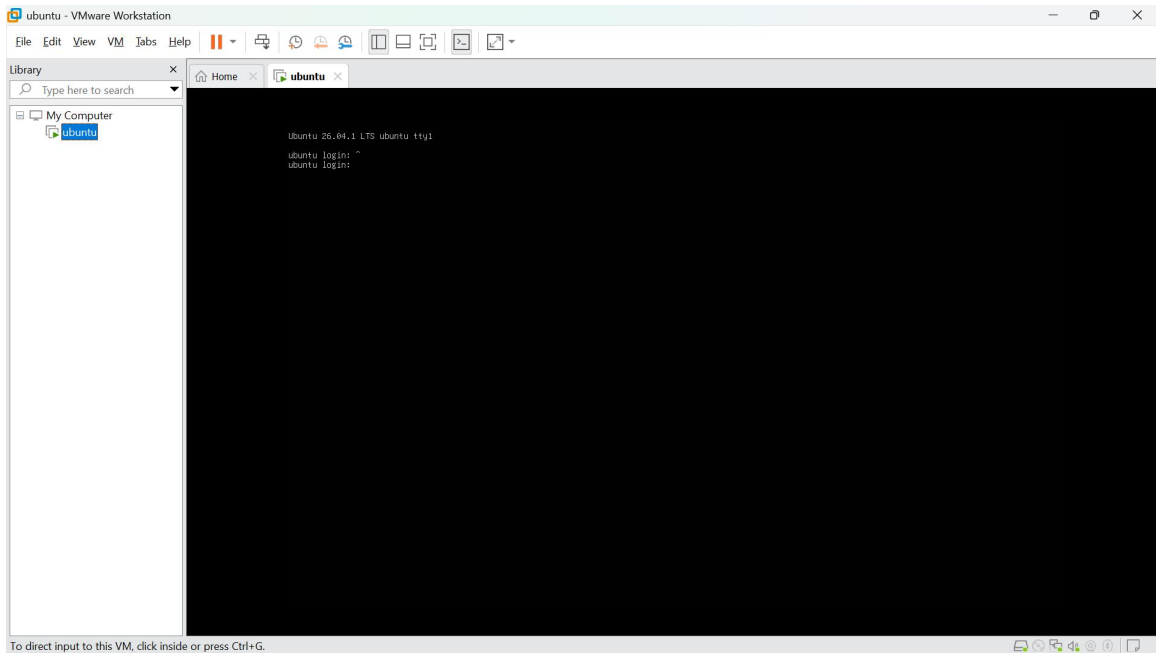


```
Installation complete! [ Help ]

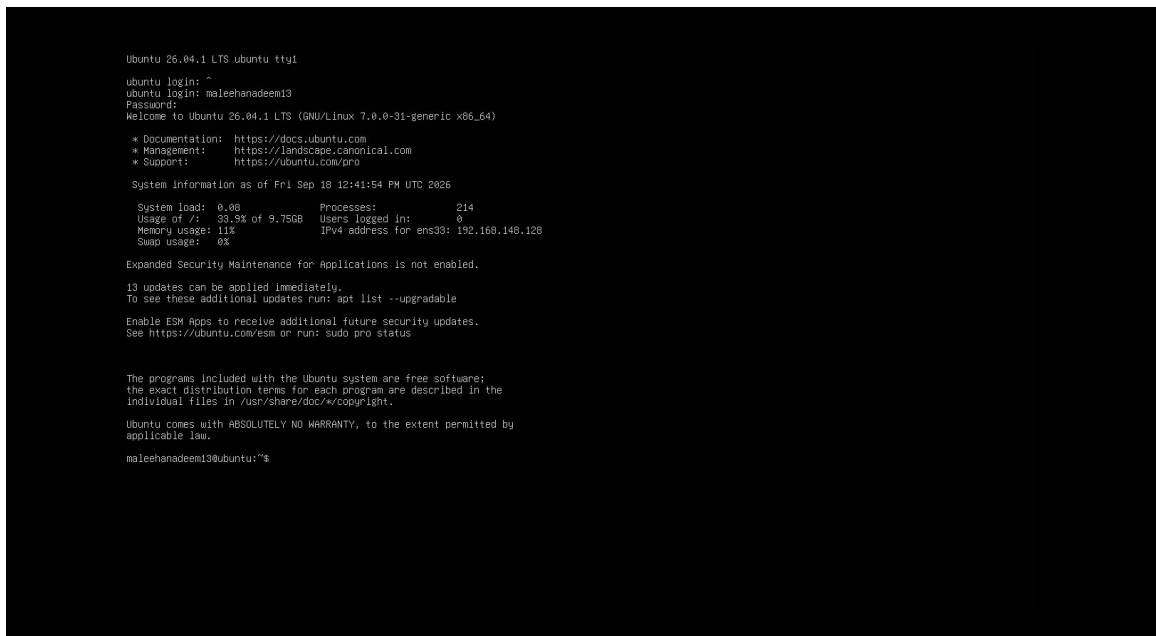
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmp1pvc14r/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
curtin command in-target
curtin command in-target
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/run_builtin:
subiquity/late/run_user_supplied:
```

[View full log](#)
[Reboot Now]

Installation-complete screen with the 'Reboot Now' option.



Ubuntu Server login screen shown after the first reboot.



Successful terminal login using the Ubuntu username and password configured in Task 13.

```
Ubuntu 26.04.1 LTS ubuntu tty1
ubuntu login: ^
ubuntu login: maleehanadeem13
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 18 12:41:54 PM UTC 2026

System load: 0.00          Processes:            214
Usage of /:  33.9% of 9.75GB Users logged in:             0
Memory usage: 11%         IPv4 address for ens33: 192.168.148.128
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

maleehanadeem13@ubuntu:~$
maleehanadeem13@ubuntu:~$
maleehanadeem13@ubuntu:~$
maleehanadeem13@ubuntu:~$ _
```

Terminal prompt confirming the identity format maleehanadeem13@ubuntu:~\$, matching the required <github-username>@ubuntu pattern.

Task 16: Find the Ubuntu Server IP Address

```
maleehanadeem13@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:35:9b:1d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29359b1d
    inet 192.168.148.128/24 metric 100 brd 192.168.148.255 scope global dynamic ens33
        valid_lft 1424sec preferred_lft 1424sec
    inet6 fe80::20c:29ff:fe35:9b1d/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
maleehanadeem13@ubuntu:~$
```

Output of the 'ip addr' command showing the ens33 interface with IPv4 address 192.168.148.128, distinct from the 127.0.0.1 loopback address.

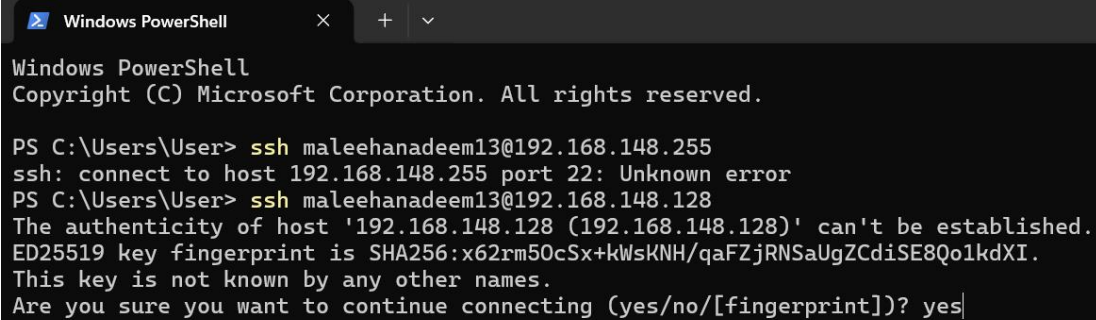
```
altname enp2s1
altname enx000c29359b1d
inet 192.168.148.128/24 metric 100 brd 192.168.148.255 scope global dynamic ens33
    valid_lft 1424sec preferred_lft 1424sec
```

Close-up confirmation of the assigned IPv4 address 192.168.148.128 on interface ens33, distinct from the loopback address.

Task 17: Connect to Ubuntu Server from Windows

```
PS C:\Users\User> ssh maleehanadeem13@192.168.148.128
The authenticity of host '192.168.148.128 (192.168.148.128)' can't be established.
ED25519 key fingerprint is SHA256:x62rm50cSx+kWsKNH/qaFZjRNSaUgZCdiSE8Qo1kdXI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

SSH command run from Windows to connect to the Ubuntu Server using its IPv4 address (192.168.148.128).



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\User> ssh maleehanadeem13@192.168.148.255
ssh: connect to host 192.168.148.255 port 22: Unknown error
PS C:\Users\User> ssh maleehanadeem13@192.168.148.128
The authenticity of host '192.168.148.128 (192.168.148.128)' can't be established.
ED25519 key fingerprint is SHA256:x62rm50cSx+kWsKNH/qaFZjRNSaUgZCdiSE8Qo1kdXI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes|
```

Server fingerprint prompt accepted by typing 'yes' during the first SSH connection.

```
maleehanaadeem13@ubuntu: ~$ ssh maleehanaadeem13@192.168.148.255
ssh: connect to host 192.168.148.255 port 22: Unknown error
PS C:\Users\User> ssh maleehanaadeem13@192.168.148.128
The authenticity of host '192.168.148.128 (192.168.148.128)' can't be established.
ED25519 key fingerprint is SHA256:x62rm50cSx+kWskNH/qaFZjRNSaUgZCdiSE8Qo1kdXI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.148.128' (ED25519) to the list of known hosts.
Connection closed by 192.168.148.128 port 22
PS C:\Users\User> ssh maleehanaadeem13@192.168.148.128
maleehanaadeem13@192.168.148.128's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 18 01:17:45 PM UTC 2026

System load:  0.09          Processes:      223
Usage of /:   34.4% of 9.75GB Users logged in:   0
Memory usage: 12%          IPv4 address for ens33: 192.168.148.128
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

maleehanaadeem13@ubuntu:~$
```

Successful SSH login into the Ubuntu Server from Windows.

```
maleehanaadeem13@ubuntu: ~$
Memory usage: 12%          IPv4 address for ens33: 192.168.148.128
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

maleehanaadeem13@ubuntu:~$
maleehanaadeem13@ubuntu:~$
maleehanaadeem13@ubuntu:~$

maleehanaadeem13@ubuntu:~$
```

Remote SSH terminal prompt confirming maleehanaadeem13@ubuntu:~\$, matching the required identity pattern.

